DM
DEPARTAMENTO
DE MATEMÁTICA
TÉCNICO LISBOA

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Project 1: Dimension Reduction in Classification

Mathematics for Machine Learning

(MECD & MMAC, 2nd Semester, 2025/2026)

Handed out on March 16, 2026.

Intermediate presentation: March 30, 2026.

To be handed back by April 19, 2026.

This project should be done in R. Start by installing R followed by RStudio:

(i) Install the R program (`http://www.r-project.org`). In the download section choose your operating system (Mac, Linux or Windows) and what is your 32- or 64-bit.

(ii) Install RStudio (`https://www.rstudio.com/products/rstudio/download/`).

Consider the dataset associated description available at: Prediction of music genre (see [3] for additional information).

1. Consider only two genders: Classical and Rap. Make a preliminary analysis of the data, identify potential problems. Discuss your findings.
2. Fix a seed, and randomly divide your dataset into 70% for training and 30% for testing.
3. Perform Principal Component Analysis (PCA) on the numerical features. Determine how many components to retain and interpret the first two principal components. Justify any feature transformations used in your analysis.
4. Perform Kernel Principal Component Analysis (Kernel PCA) on the numerical features. Decide how many components to retain. Explore different kernel functions and hyperparameter settings..
5. Using the training dataset and the available input features, apply the following information-theoretic forward feature selection methods to identify the features most relevant to the classification task. (see [1,5,6]): MIM, MIFS, mRMR, maxMIFS, CIFE, CMIM, JMI, DMIM. Decide about the number of features to be retain.
6. Consider two classifiers: k-Nearest Neighbour (kNN) with $k = 5$ neighbours or another classifier at your own choice (see [2,4,6]).
   (a) Train the classifiers using (i) the original dataset and (ii) the reduced representations obtained from the dimensionality reduction methods discussed in questions 3, 4, and 5.

(b) For each case, predict the class of each test set observation.

(c) Compare the true class and the assigned class for the test set, and estimate the following measure of performance (see [3,4]):

i. Accuracy (Acc): percentage of corrected assigned observations.

ii. Macro_Recall (Macro_Re): arithmetic mean of the classes recall, where the recall of the i-th class ($Re(i)$) is the percentage of the observations of the i-th class correctly assigned to that class.

iii. Macro_Precision (Macro_Pr): arithmetic mean of the classes precision, where the precision of the i-th class ($Pr(i)$) is the percentage of observation assigned to the i-th class that truly belong to the i-th class.

iv. Macro_F_1 measure (Macro_F_1): arithmetic mean of the classes F_1-measure, where the F_1-measure of the i-th class is the harmonic mean of the i-th class recall and precision.

7. Compare and discuss the results resulting from the different strategies. Include in your discussion all options that you have made, the advantages, and disadvantages of each alternative.

References:

[1] G. Brown, A. Pocock, M.-J. Zhao, M. Luján. Conditional likelihood maximisation: A unifying framework for information theoretic feature selection, *J. Mach. Learn. Res.*, 13, 27–66, 2012.

[2] G. James, D. Witten, T. Hastie, and R. Tibshirani. *An Introduction to Statistical Learning with Applications in R*. Springer, NY, 2013 (Printing 2021).

[3] Prediction of music genre [Dataset]. Kaggle Repository. https://www.kaggle.com/datasets/vicsuperman/prediction-of-music-genre/data

[4] M. Kuhn. Building Predictive Models in R Using the caret Package. *Journal of Statistical Software*, 28(5), 1–26, 2008. `doi:http://dx.doi.org/10.18637/jss.v028.i05`. See A Short Introduction to the caret Package or The caret Package.

[5] F. Macedo, R. Valadas, E. Carrasquinha, M. R.o Oliveira, A. Pacheco, Feature selection using Decomposed Mutual Information Maximization, Neurocomputing, 513, 2022, 215-232, 2022.

[6] C. Pascoal, M. R. Oliveira, A. Pacheco, and R. Valadas. Theoretical evaluation of feature selection methods based on mutual information, *Neurocomputing*, 226(1) 168–181, 2017.

[7] H. Zhou, X. Wang, R. Zhu. Feature selection based on mutual information with correlation coefficient, *Applied Intelligence*, 52(5), 5457–5474, 2022.

- **Progress report:**
  - On **30 March** 2026, each group should do a 3 min presentation, reporting their progress, difficulties, and future work on the Project 1.

- **About the groups:**
  - Students should organize themselves in groups of 2 persons.

- **About the report:**
  - You have to deliver a report including an explanation of what you have done, the visualization of our results, and a critical discussion of your results, together with the commented code of your project.
  - The report should not exceed 8 pages.
  - The **commented R code** and the **report** must be uploaded to the Fénix webpage.